



What's in Your Transit? Towards Reliably Getting 5× More Science from Exoplanet Transit Data

SAMSON J. MERCIER ,¹ JULIEN DE WIT ,¹ AND BENJAMIN V. RACKHAM ^{1,2}

¹*Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, MA 02139, USA*

²*Kavli Institute for Astrophysics and Space Research, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*

ABSTRACT

Exoplanetary science heavily relies on transit depth (D) measurements. Yet, as instrumental precision increases, the uncertainty on D drifts from expectations driven solely by photon-noise. Here, we characterize this shortfall (the Transit-Depth Precision Problem) by defining an amplification factor, A , quantifying the discrepancy between measured transit-depth uncertainty and baseline scatter. While in theory A should be $\sim \sqrt{3}$, we find that it departs substantially from this limit due to correlations between D and the limb-darkening coefficients (LDCs). Grounding our analyses with over 500,000 intensity profiles from modern stellar atmosphere models spanning a wide range of T_{eff} , $\log g$, $[M/H]$, and wavelength, we show that reliable LDC priors can substantially improve transit-based science, while low-fidelity ones can instead bias D . The prevailing practice of using a quadratic limb-darkening law yields low $A \sim 3$ –5 but transit depth biases of 5–10 σ —potentially introducing spurious spectral features affecting atmospheric studies. These effects are most severe at short wavelengths and for host stars with low metallicity or low effective temperature, where the underlying intensity profile is strongly non-quadratic. As a near-term mitigation, we recommend fitting transits with both a third-order polynomial law *and* a fourth-order non-linear law which optimally balance bias and A while cross-checking one another's fidelity. With reasonable LDC priors (10–20% width) A increases only mildly to ~ 5 –6, while bias drops below 2σ . The long-term solution lies in improving the stellar atmosphere models and transit-fitting tools to reduce A and bias jointly, thus enabling the same science with up to $\sim 5\times$ fewer transits.

Keywords: Transmission spectroscopy (2133); Stellar atmospheres (1584); Planet hosting stars (1242); Exoplanet atmospheres (487); Fundamental parameters of stars (555)

1. INTRODUCTION

Since its launch, the James Webb Space Telescope (JWST) has ushered in a new era of exoplanet science. Thanks to the unprecedented precision of its data, we can probe in finer detail the structure and composition of exoplanet atmospheres (*e.g.*, Xue et al. 2024; Welbanks et al. 2024) and investigate novel processes such as photochemistry (*e.g.*, Tsai et al. 2023; Dyrek et al. 2024), atmosphere–interior interactions (*e.g.*, Tsai et al. 2021; Benneke et al. 2024), and planetary oblateness (*i.e.*, internal processes, *e.g.*, Berardo & de Wit 2022; Liu et al. 2025).

Although instrumentation has improved drastically over the last decades, the models and frameworks used to interpret data have lagged behind. One example is our incomplete understanding of stars: stellar magnetic active regions (spots and faculae) consistently contaminate photometric and spectroscopic observations (*e.g.*, Dumusque et al. 2011b,a; Meunier et al. 2017; Rackham et al. 2017, 2018, 2019, 2023; Saba et al. 2025). Our inability to model or mitigate these effects leads to biased measurements of the radial velocity amplitude and transit depth from which the most basic exoplanet properties—mass and radius, respectively—are measured (*e.g.*, Rabus et al. 2009; Carter et al. 2011; Lovis et al. 2011). Similar biases can arise from other model limitations, such as with line shape parameterizations in opacity models (Niraula et al. 2022, 2023). These model limitations propagate throughout our anal-

yses and hamper our ability to draw meaningful conclusions on exoplanet characteristics.

In this context, here we examine the information content of exoplanet transit data with the goal of identifying what targeted improvements to methods and models can enhance the scientific return of these observations. In Section 2 we summarize prior work on limb-darkening in the context of exoplanet transits and clarify the novelties that our study adds to this well-established body of work. In Section 3, we present the root cause for our investigation, the “Transit-Depth Precision Problem” (TDPP). In Section 4 we construct the injection–retrieval framework underlying our subsequent analyses, grounding our choice of limb-darkening parametrization in the statistics of modern stellar models. In Section 5, we use this framework to investigate the origins of the TDPP and quantify their respective contributions. In Section 6 we discuss targeted mitigation strategies and quantify the substantial boost in scientific return the community could expect from their adoption. We conclude in Section 7. The whole paper is reproducible and can be found by clicking the GitHub icon .

2. LIMB-DARKENING: PRIOR WORK AND OPEN CHALLENGES

Limb-darkening, or center-to-limb intensity variation, occurs because radiation emerging near the stellar limb originates from higher, cooler atmospheric layers and is therefore weaker than radiation from the disk center. Since the advent of exoplanet transit observations (Charbonneau et al. 2000; Henry et al. 2000), limb-darkening has been recognized as a critical component of transit modeling, as it directly shapes the morphology of transit light curves.

Limb-darkening is typically described with parametric limb-darkening laws (LDLs), which express the stellar intensity as a function of the viewing angle μ^1 through a set of limb-darkening coefficients (LDCs). A wide variety of LDLs have been proposed, ranging from linear (Milne 1921) and quadratic forms (Manduca et al. 1977) to higher-order polynomials (Giménez 2006) and non-linear prescriptions (Claret 2000; Sing 2010). In parallel, methodological advances have focused on reparameterizing LDLs to reduce parameter correlations (Maxted 2018, 2023) and improve the sampling of LDC parameter spaces (Kipping 2013, 2016).

This diversity of approaches reflects a fundamental tension between the intrinsic complexity of stellar intensity profiles, the increasing precision of transit obser-

vations, and the need for analytically and numerically tractable models. Stellar atmosphere models have long shown that realistic intensity profiles often require some degree of non-linearity for accurate representation (*e.g.*, Claret 1998; Neilson & Lester 2013a,b). Nevertheless, low-order LDLs remain widely used due to their simplicity and low dimensionality, which facilitate inference.

An extensive body of work has demonstrated that such simplified LDLs lead to bias and increased variance in transit depth measurements (*e.g.*, Southworth 2008; Carter et al. 2008; Csizmadia et al. 2013; Neilson & Lester 2013a,b; Espinoza & Jordán 2015, 2016; Morello et al. 2017; Keers et al. 2024). These effects arise from strong correlations between limb-darkening parameters and other transit parameters, which fundamentally limit the information that can be extracted from transit light curves. Such limitations become increasingly relevant for high-quality transit data, such as those acquired with JWST, where photon noise is low enough that modelling degeneracies dominate the error budget. Moreover, degraded transit depth precision and accuracy have been observed across a wide range of orbital geometries, wavelength regimes, and stellar types, underscoring that the choice of LDL has a ubiquitous impact on transit analyses. Many studies therefore recommend the use of higher-order laws, such as the three-parameter prescription of Sing (2010), to mitigate both bias and inflated variance. However, while higher-order LDLs reduce these effects, they do not eliminate them entirely, and the optimal choice of LDL remains strongly dependent on data quality (see Figure 7 of Espinoza & Jordán 2016, hereafter EJ16).

In summary, previous works have characterized limb-darkening effects in terms of bias, covariance structure, or goodness-of-fit, and explicitly quantified the information content of transit depth measurements relative to the photon-noise limit. However, no study has adopted a holistic MCMC-based approach to jointly analyze all fitted transit parameters and quantify their individual contributions to transit depth uncertainty through the structure of the full, multi-dimensional posterior. In this work, we address this gap and frame limb-darkening as an information bottleneck. Through injection–retrievals we quantitatively explore how transit depth precision and bias depend on the adopted limb-darkening prescription and prior assumptions. To ensure our conclusions are not tied to a particular stellar or wavelength regime, we further ground our framework with intensity profiles extracted from modern stellar atmosphere models spanning a wide range of T_{eff} , $\log g$, $[M/H]$, and λ . Building on this setup, we quantify the sensitivity of the transit depth to all fitted transit parameters, iden-

¹ $\mu = \cos(\theta)$ where θ is the angle between the observer’s line of sight and the stellar surface normal.

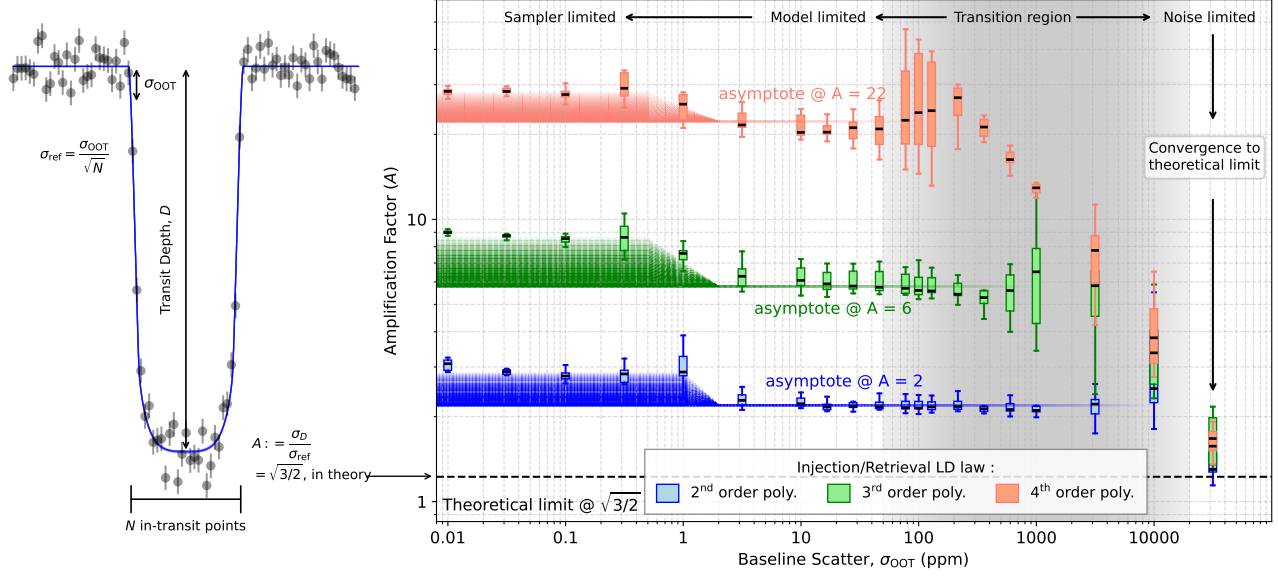


Figure 1. Expectations vs reality in constraining an exoplanet transit depth. *Left:* Mock transit generated with the properties listed in Table B1. Black points show the light curve with 359 parts per million (ppm) of injected Gaussian white noise. Blue curves are model realizations drawn from the posterior distribution of D obtained by fitting this dataset. Annotated arrows indicate the quantities relevant to the TDPP. *Right:* Illustration of the TDPP. Each colored box plot set corresponds to injection–retrieval tests done with a different limb-darkening prescription. For each set, the injection and retrieval models have the same functional form. Box plots summarize ten MCMC retrievals with identical scatter but different noise realizations. For each box, the black horizontal line marks the median, the lower and upper edges the first and third quartiles, and whiskers the minimum and maximum values (calculated as $1.5\times$ the first and third quartile). The four annotated regimes are discussed in Section 3. Colored horizontal lines show the model-limited asymptote for each injection–retrieval family, fading toward the transition region. The injected out-of-transit scatter (σ_{OOT}) is measured in 120-min bins.

tify which correlations dominate information loss, and assess what complementary data are most effective at mitigating the TDPP.

3. THE TRANSIT-DEPTH PRECISION PROBLEM

The most informative observable for atmospheric science across the board is the transit depth, $D(\lambda) = (R_p(\lambda)/R_\star(\lambda))^2$, in which $R_p(\lambda)$ is the planetary radius and $R_\star(\lambda)$ is the stellar radius. Measuring D at different wavelengths reveals the planet’s apparent size as a function of wavelength, which can be translated into information on atmospheric opacity (*e.g.*, Seager & Sasselov 2000; Benneke & Seager 2012; de Wit & Seager 2013). The precision with which D is measured directly sets the limit on the inferences that can be made about atmospheric characteristics (*e.g.*, Deming et al. 2009; Greene et al. 2016). It is thus crucial that the uncertainty on the transit depth (σ_D) be minimized—as well as biases on it, naturally. Since variations in transit depth linked to atmospheric opacity are notably weaker than those linked to the TDPP, we assume hereafter that D is wavelength-independent.

To first order, one could expect that σ_D relates to the scatter in the photometric time series (Figure 1, left panel), which includes both photon noise (roughly Gaussian at high count levels) and correlated noise (marginal

to first order in many modern datasets), measurable as the out-of-transit scatter for a bin of duration similar to the transit duration ($\sigma_{\text{ref}} = \sigma_{\text{OOT}}/\sqrt{N}$ with N in-transit points). Specifically, one expects that $\sigma_D = \sqrt{3} \sigma_{\text{ref}}$, as the typical out-of-transit baseline for transit observations is twice the duration of the transit event targeted (see Appendix A for a full derivation). However, when comparing these two quantities in many existing datasets, a significant discrepancy between expectation and reality emerges (*i.e.*, $\sigma_D/\sigma_{\text{ref}} \gg \sqrt{3}$). To simplify notation in our subsequent investigations, we define the amplification factor $A = \sigma_D/\sigma_{\text{ref}}$.

We illustrate this discrepancy through a set of injection–retrieval tests, whose results are shown in the right panel of Figure 1. For these tests, we simulated a single transit with the `jaxoplanet` package (Hattori et al. 2024) for a hypothetical system (see Table B1 for its properties) and varying levels of white noise, before fitting the mock data with the same transit model. We consider three families of tests, each using a different LDL for both injection and retrieval: a second-, third-, and fourth-order polynomial LDL, respectively. These three families are intended to illustrate how the dimensionality of the limb-darkening parameterization affects the TDPP, with each additional order introducing one

more free parameter. The transit is constructed with equal pre-, post- and in-transit durations (as in Figure 1, left panel). To sample the posterior, we used the affine invariant Markov Chain Monte Carlo (MCMC) method implemented in the `emcee-jax` package (a JAX implementation of `emcee`, Foreman-Mackey et al. 2013) with 50 walkers run for 100,000 steps, discarding the first 70,000 as burn-in. We applied uninformative uniform priors to all fitted parameters except the orbital period, on which we set a strong Gaussian prior, and the transit midpoint, which was fixed. This setup reflects the fact that such analyses generally assume prior observations have tightly constrained transit timings. Walkers were initialized in a wide multi-dimensional Gaussian ball around truth. Poorly converged samples are excluded via three rounds of iterative inter-quartile range (IQR) clipping². From D 's posterior we measured σ_D , and with the injected white noise level we computed the corresponding A . All details regarding the retrieval setup, including the LDC values used for injection in each of the three cases, can be found in Table B1.

Each colored box plot set in the right panel of Figure 1 highlights one of the injection–retrieval families. The behavior of these sets falls into four regimes. At high scatter, the amplification factor converges to the theoretical limit of $\sqrt{3}$, indicating that the retrievals are noise-limited. As scatter decreases, a transition region is crossed wherein retrievals shift from being noise- to model-limited (shown as a shaded band). In the model-limited regime, A plateaus at a value set by the degeneracies and correlations among the fitted model parameters. Indeed, when fitting transits, parameters describing the exoplanet's orbit and the stellar intensity profile must be inferred alongside the transit depth. Many of these affect the transit shape in similar ways, producing compensatory effects where different parameter combinations yield similar light curves (see, e.g., Figure 2 and de Wit et al. 2012, Fig. 4). Such dependence broadens the posterior distribution of D , increasing A . The plateau value depends on the dimensionality of the retrieval: as more model parameters are included, for example when moving from the second- to fourth-order polynomial LDL, the space of iso-probability solutions expands, thus broadening the posterior. Finally, at extremely low scatter, the likelihood landscape becomes so sharply peaked that the sampler struggles to explore it efficiently, and A is no longer limited by the model but by the ability of the MCMC to draw independent

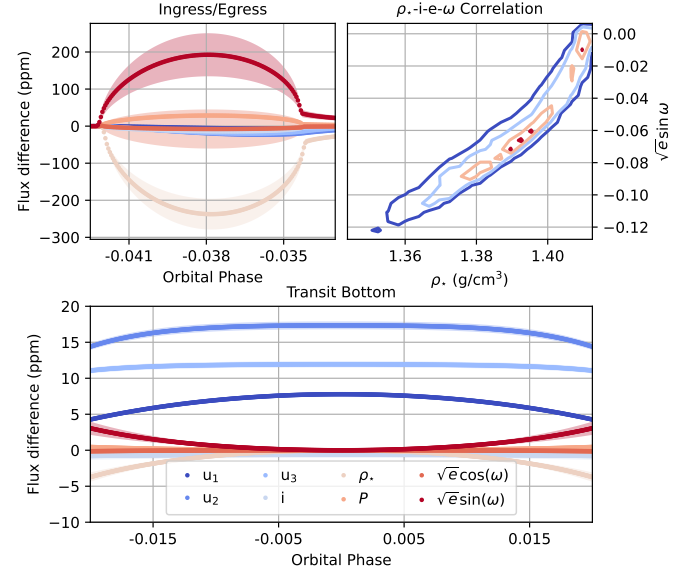


Figure 2. Parameter-induced perturbations on transit shape. Separate panels highlight the transit ingress/egress, and bottom. Symmetric “bumps” can be clearly seen at ingress and egress, driven by the orbital parameters. In contrast, fluctuations in the bottom are dominated by the LDCs. Perturbations beyond the photometric precision (16 ppm here) are due to underlying correlation, as exemplified in the top-right panel by the anti-correlation between ρ_* and $\sqrt{e}\sin\omega$. Colored contours indicate the 10, 30, 60, and 90% iso-probability levels. In practice, such correlations couple all orbital parameters simultaneously.

samples. This sampler-limited regime is indicated by the upward fade of the asymptote lines towards the left.

4. AN INFORMED INJECTION–RETRIEVAL FRAMEWORK

With the existence of the TDPP established (which holds regardless of the specific system or LDC values used) we now turn to diagnosing it quantitatively. Before doing so, we must specify the injection–retrieval framework underlying the subsequent analyses. As Section 3 makes clear, limb-darkening is at the core of the TDPP, thus the related choices we make directly shape what our analyses can reveal.

In particular, two choices are required: the LDL used for injection and retrieval, and the LDCs used for injection. Because our test system is hypothetical, no literature LDCs exist and making arbitrary choices could limit the generalizability of our findings. We therefore ground both choices—particularly the latter—in modern stellar model grids, ensuring that our adopted setup spans representative regions of the relevant physical space.

We note that all injection–retrievals throughout this work assume a central transit geometry ($b = 0$). Grazing

² Clipping is applied to individual (walker, step) pairs rather than to individual chains. At each round, pairs whose χ^2 or parameter values fall outside median $\pm \{5, 4, 3\} \times \text{IQR}$ are removed.

and near-grazing configurations introduce qualitatively different parameter degeneracies, and dedicated follow-up work varying the orbital properties of the hypothetical system—including its impact parameter—will be performed to determine how our recommendations generalize to those regimes.

4.1. Selecting an appropriate law

All light curves generated for this section’s analyses were produced using the fourth-order non-linear LDL

$$\frac{I(\mu)}{I(1)} = 1 - \sum_{k=1}^4 c_k (1 - \mu^{k/2}) \quad (1)$$

with $I(\mu)$ the intensity at a given position on the stellar disk. This prescription has been shown to produce the most accurate representation of stellar intensity profiles and currently appears to be the most appropriate choice for high-precision transit observations (*e.g.*, [EJ16](#); [Morello et al. 2017](#); [Keers et al. 2024](#)). For the retrieval law, we adopt a third-order polynomial LDL:

$$\frac{I(\mu)}{I(1)} = 1 - \sum_{k=1}^3 u_k (1 - \mu)^k. \quad (2)$$

We demonstrate in [Section 6](#) this provides the best compromise between flexibility, bias, and amplification factor at the noise level considered. Hereafter, (c_1, c_2, c_3, c_4) refers to the coefficients of the fourth-order non-linear LDL and (u_1, u_2, u_3) to those of the third-order polynomial LDL.

4.2. Selecting appropriate coefficients

Rather than selecting an arbitrary set of LDCs, we extract representative vectors (c_1, c_2, c_3, c_4) from the population of physically plausible intensity profiles spanning the full range of stellar parameters and wavelengths available in stellar models. This ensures both the physical fidelity of our injection and a means of assessing how sensitive our conclusions are to the physical inputs.

Profile generation. Using `ExoTic-LD` ([Grant & Wakeford 2024](#)), we extract stellar intensity profiles from the MPS-ATLAS set1 stellar models ([Kostogryz et al. 2022](#)) on a $10 \times 10 \times 10$ grid, uniformly sampling $T_{\text{eff}} \in [3500, 9000]$ K, $\log g \in [3.0, 5.0]$, and $[M/H] \in [-5.0, 1.5]$. Profiles are available at 1221 wavelengths ($\lambda \in [0.0099, 160]$ μm) per grid point, yielding a total dataset size of 1,221,000. We adopt MPS-ATLAS set1 as it is among the most up-to-date set of stellar models and yields LDCs consistent with its set2 counterpart ([Kostogryz et al. 2023](#)). Restricting ourselves to only MPS-ATLAS set1 avoids introducing inter-model variability unrelated to our driving question. This choice

also defines the parameter ranges explored, as different models cover differently the $(T_{\text{eff}}, \log g, [M/H], \lambda)$ -space.

Filtering. We restrict the wavelength coverage to the JWST NIRSpec PRISM range (0.6–5.3 μm) for two reasons: it targets a broad wavelength region relevant for most JWST transit observations, and it removes the grid boundaries where numerical artifacts produce non-monotonic intensity profiles³. After this cut, 585,000 profiles remain. We then manually mask any residual non-monotonic profiles, removing an additional 441 ($<0.1\%$). These preferentially originate from high wavelengths (2.52–5.29 μm) and from low T_{eff} , $\log g$, and $[M/H]$ —regimes where limb-darkening is intrinsically weak, either because the plane-parallel approximation breaks down (low $\log g$) or because atmospheric complexity increases (*e.g.*, molecular opacities at low T_{eff} , or H^- continuum dominating at low $[M/H]$). In both cases, the center-to-limb intensity contrast is sufficiently reduced that numerical noise can produce spurious profile inversions. A detailed examination of these artifacts and their dependence on stellar parameters is left to future work. Our final dataset comprises 584,559 profiles.

Coefficient extraction. Each profile is interpolated onto a uniform 100-point μ -grid via a cubic spline and fit with the fourth-order non-linear LDL⁴. This approach to intensity profile fitting, first presented by [Claret & Bloemen 2011](#), was shown by [Espinoza & Jordán 2015](#) (hereafter [EJ15](#)) to minimize the residuals between the true profile and its parametric approximation. In doing so, we reduce the dataset from a $(T_{\text{eff}}, \log g, [M/H], \lambda)$ -dependent intensity field to a four-dimensional space of LDC vectors (c_1, c_2, c_3, c_4) . The resulting distribution is shown in [Figure C1](#) and [Figure C2](#), where the clear structure reflects the inherited dependence on the underlying physical parameters. A detailed exploration of this topology is left to future work.

Clustering and mode identification. To extract representative LDC vectors from this dataset, we apply hierarchical agglomerative clustering ([Everitt et al. 2011](#)) on a randomly drawn subset of 50,000 vectors (sufficient to preserve the overall structure while keeping the pairwise-distance computation tractable). We used the Mahalanobis metric with Ward linkage ([Ward 1963](#)) and a distance cutoff of 80, all selected empirically by visu-

³ At extreme wavelengths, intensity values become very small causing numerical precision losses that allow intensities at adjacent μ values to overlap spuriously. For instance, $I(\mu = 0) < I(\mu = 0.25)$, which is nonphysical since intensity decreases from the disk center outwards.

⁴ Throughout this work, all LDL fits to an intensity profile are performed via non-linear least squares optimization using `lmfit` ([Newville et al. 2016](#)).

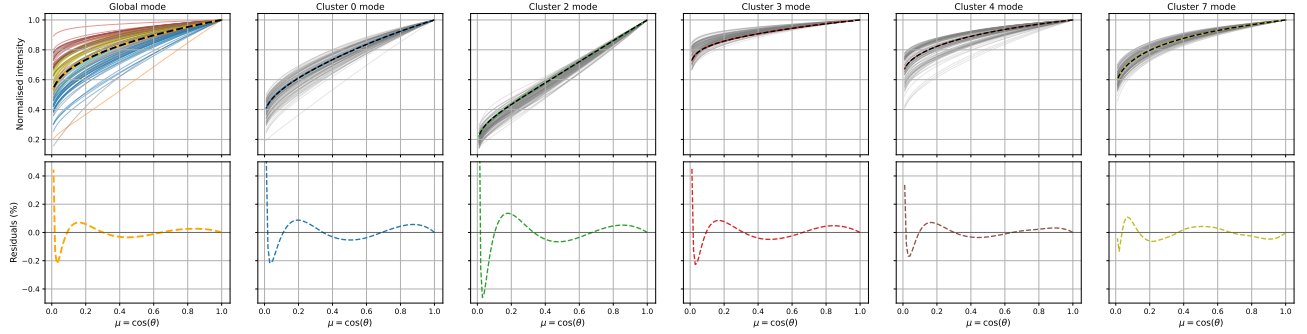


Figure 3. Representative intensity profiles. Intensity profiles for the global and cluster modes introduced in Section 4 and used in the analyses of Section 5 and Section 6. *Top:* A random subset of 100 profiles is shown in each panel’s background: for the global mode panel, profiles are colored by their cluster assignment (see Figure C3); for each cluster mode panel, profiles belonging to that cluster are shown in gray. The modal intensity profile for each panel is overlaid as a solid colored line, with its fourth-order non-linear LDL fit shown as a dashed black line. The resulting LDCs along with the stellar parameters and wavelength used to generate each profile are reported in Table B1. *Bottom:* Relative difference residuals between the fourth-order non-linear LDL fit and the true intensity profile. These clearly remain well below the 1% level, with the exception of the cluster 4 mode where residuals grow towards the low- μ tail as the intensity profile’s behavior becomes increasingly complex.

ally inspecting the resulting clustering across a range of combinations. Singleton clusters are reassigned to the nearest non-singleton cluster. Our approach yields nine clusters, whose membership is shown in Figure C3. For each of them, we identify the LDC vector closest to its centroid as the “cluster mode.” The vector closest to the global centroid of all points is designated as the “global mode,” representing the single most typical profile across the physical space explored. The global mode is used as the injection vector for the analyses in Section 5. In Section 6, we inject with the modes of three clusters (clusters 1, 3, and 4; see Figure 3), testing how our conclusions generalize across the $(T_{\text{eff}}, \log g, [\text{M}/\text{H}], \lambda)$ -space. As shown in Figure C3, we select these clusters as they roughly cover the dataset edges and center. The LDC vectors and corresponding physical parameters for all four modes are reported in Table B1, and span a diverse range of stellar types and observing wavelengths:

- **Global mode** — a near-solar metallicity K/M-type at mid-infrared wavelengths (< 4500 K, 3.44, 0.06, $3.21\mu\text{m}$)
- **Cluster 1** — a metal-poor K/M-type in the near-infrared (< 4500 K, 3.22, -4.28 , $1.07\mu\text{m}$)
- **Cluster 3** — a moderately metal-poor G-type star in the optical (5333 K, 3.44, -2.11 , $0.88\mu\text{m}$)
- **Cluster 4** — a metal-rich K-type in the optical (4722 K, 3.22, 1.50, $0.93\mu\text{m}$).

5. DIAGNOSING THE TDPP

Equipped with an injection–retrieval framework, we now investigate what drives the increase in A beyond the noise-limited regime ($\sigma_{\text{OOT}} \lesssim 30,000$ ppm). As Section 3 makes clear, this amplification originates from

degeneracies between model parameters. Here we identify which parameters are responsible and quantify their relative contributions.

5.1. Perturbation analysis

The model-driven increase in A results from correlations between transit model parameters. Different combinations can generate nearly identical light curves, making it difficult to disentangle each parameter’s contribution and recover its true value. A simple way to illustrate this effect is to perturb each parameter by 1σ (measured from the posterior output by a MCMC) and examine how the light curve responds, an approach previously used by de Wit et al. (2012) to study secondary eclipses. We present the results of this perturbation analysis in Figure 2, specifically focusing on changes in the transit ingress/egress (top left panel), and center (bottom panel).

At ingress and egress, perturbations generate symmetric bumps, with orbital parameters producing the largest changes. This is expected: fixing the transit midpoint and varying the orbital parameters is equivalent to altering the transit duration (Seager & Mallén-Ornelas 2003). Parameters that induce similar amplitude bumps but of opposite signs, like the stellar density (ρ_*) and $\sqrt{e} \sin \omega$, are anti-correlated (illustrated in Figure 2 top right panel). These relationships complicate posterior sampling, as algorithms like MCMC or nested sampling must navigate a landscape filled with local minima. In contrast, LDCs have little to no impact on the transit wings; variations in the LDCs impact the intensity profile but not the transit duration.

The picture is different in the transit center/bottom. Orbital parameters induce modest changes ($\sim 0\text{--}3$ ppm), whereas perturbations to the LDCs shift the bottom by

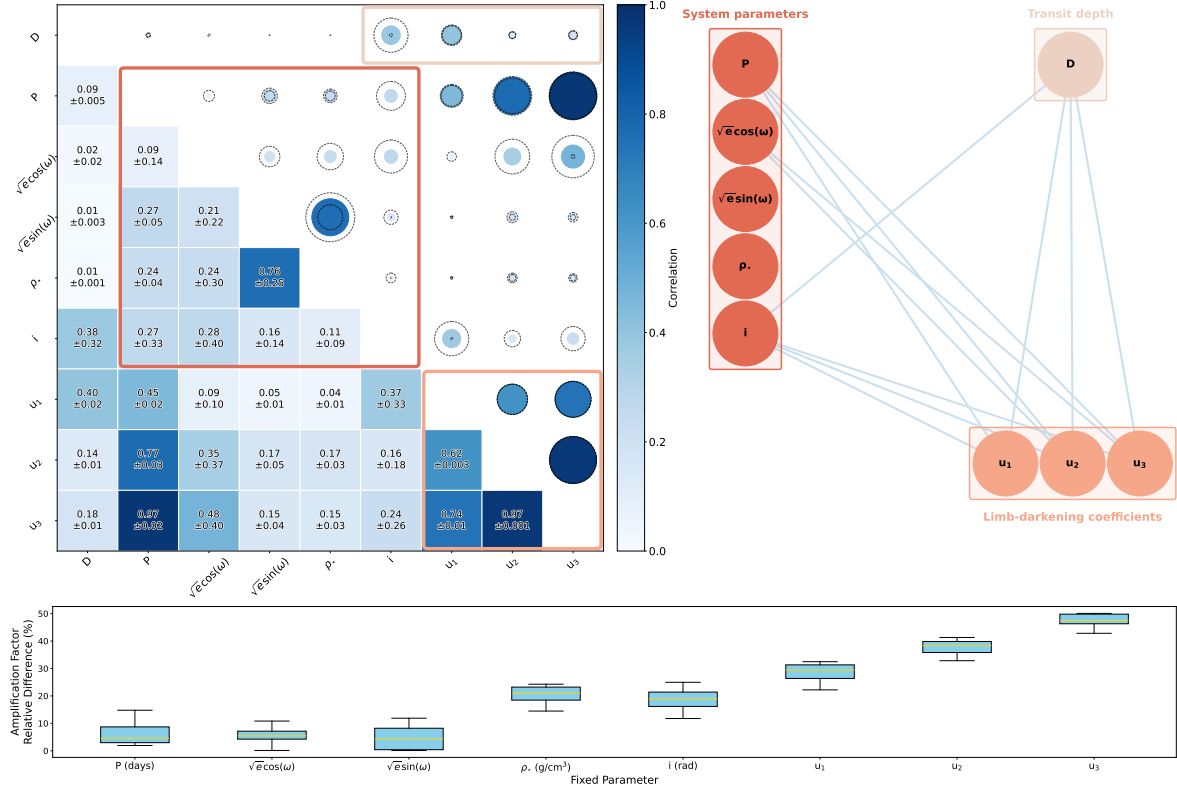


Figure 4. Revealing the underlying network of correlations. *Left:* Correlation matrix resulting from the sensitivity analysis. The lower triangle reports correlation values derived from χ^2 maps, with shading scaled to their absolute values. Colored boxes highlight families of related parameters. The upper triangle provides a circular representation of the same matrix, where circle size and shading scale with correlation strength, making the families easier to distinguish. The transit depth is most strongly correlated with the polynomial LDCs u_1 , u_2 , and u_3 . *Right:* Network representation of the same correlations with parameters grouped into families using the same color scheme as in the matrix. Blue lines highlight the strongest links between parameters of different families.

$\sim 5\text{--}15$ ppm. This reflects the role of limb-darkening in shaping the overall depth and curvature of the transit bottom. The three coefficients produce similar patterns, differing mostly in amplitude. Since D is measured from the transit bottom, these results reinforce the conclusion that LDCs are the main drivers of the TDPP.

5.2. Sensitivity analysis

To quantify parameter correlations and their impact on D , we carried out a targeted sensitivity analysis. The challenge here is to isolate correlations between pairs of parameters without contamination from the rest of the parameter space. This cannot be done via cuts in the full multi-dimensional posterior sampled by an MCMC, as marginalizing over the other parameters introduces biases. Instead, we evaluate the local χ^2 (or log-likelihood) surface over two chosen parameters at a time, holding all others fixed to truth.

For each pair, we generated a light curve, injected 16 ppm of white noise, placing the retrievals in the model-limited regime (see Figure 1), and computed the

χ^2 across a grid of values centered on truth. Because our injection and retrieval LDLs are different (see Section 4.1), we define “truth” in this case, as the LDC vector (u_1, u_2, u_3) obtained from fitting the injected fourth-order non-linear LDL with the retrieval’s third-order polynomial LDL.

Elliptical iso- χ^2 contours were then overlaid to each map. The ellipses’ rotation angle θ encodes the space’s orientation, which we translate into a correlation metric $c = \sin(2\theta)$. c peaks for diagonal contours (*i.e.*, strong correlation) and reaches a minimum for vertical or horizontal contours (*i.e.*, no correlation).

The results are summarized in Figure 4, and the full set of χ^2 maps is provided in Figure D4. The left panel presents the correlation matrix, from which two conclusions can be drawn. First, transit depth is most strongly correlated with the LDCs, confirming that they are the main contributor to the amplification factor and drive the TDPP. Second, two intraconnected families of correlations can be distinguished: one for orbital parameters and another for the LDCs. However, these families

are not isolated: clear correlations exist between their members, most importantly between $\sqrt{e} \cos \omega$, i , P and u_2 , u_3 —corresponding to the “ e - b - ρ -BD” (for brightness distribution) correlation introduced in de Wit et al. (2012). This mirrors what we observed in Figure 2, where correlated parameters (either within each family or across them) produced similar perturbations in transit shape. The LDCs therefore affect D through two channels: directly, via their own strong correlation with D , and indirectly, by serving as a conduit through which orbital parameters propagate their uncertainty into D . Together the perturbation and sensitivity analyses paint a consistent picture: the LDCs are at the heart of the TDPP, driving the increase in A both directly through their strong correlation with D and indirectly through the orbital parameters to which they couple. This underscores the need for tighter external constraints through improved stellar models or spectroscopic characterization.

6. RELIABLY BOOSTING TRANSIT SCIENCE

LDCs are the drivers of the TDPP. In the following section we discuss mitigation strategies for constraining these coefficients and illustrate the improvements that such strategies could have on reducing A .

6.1. Constraints from stellar models

A common practice when fitting transits is to constrain LDCs using model-based estimates. Tools such as ExoTiC-LD or ExoCTK (Bourque et al. 2021) provide LDC estimates from a set of stellar parameters and a given stellar atmospheric model. While these estimates are convenient, it is important to remain cautious about biases introduced when relying on them in transit fits. Indeed, EJ15 and EJ16 have shown that LDCs vary significantly between stellar models and Howarth 2011 further warns that model-derived LDCs fundamentally differ from those measured from photometric observations. Using model-derived LDCs without caution can therefore introduce systematic biases that depend on the stellar type and orbital configuration. EJ15 and EJ16 make two additional points: LDCs should not be fixed to model values but allowed to float during fitting; and some models, such as ATLAS9 (Kurucz 1979), better reproduce photometric limb-darkening than others, like PHOENIX ACES (Husser et al. 2013). While these studies caution against the blind use of model-derived LDCs, they also show that such outputs still provide good order-of-magnitude estimates to the values expected from photometry (Figs. 8 & 9 of EJ15).

Motivated by these insights, we test how varying the number of free LDCs and the strength of priors imposed

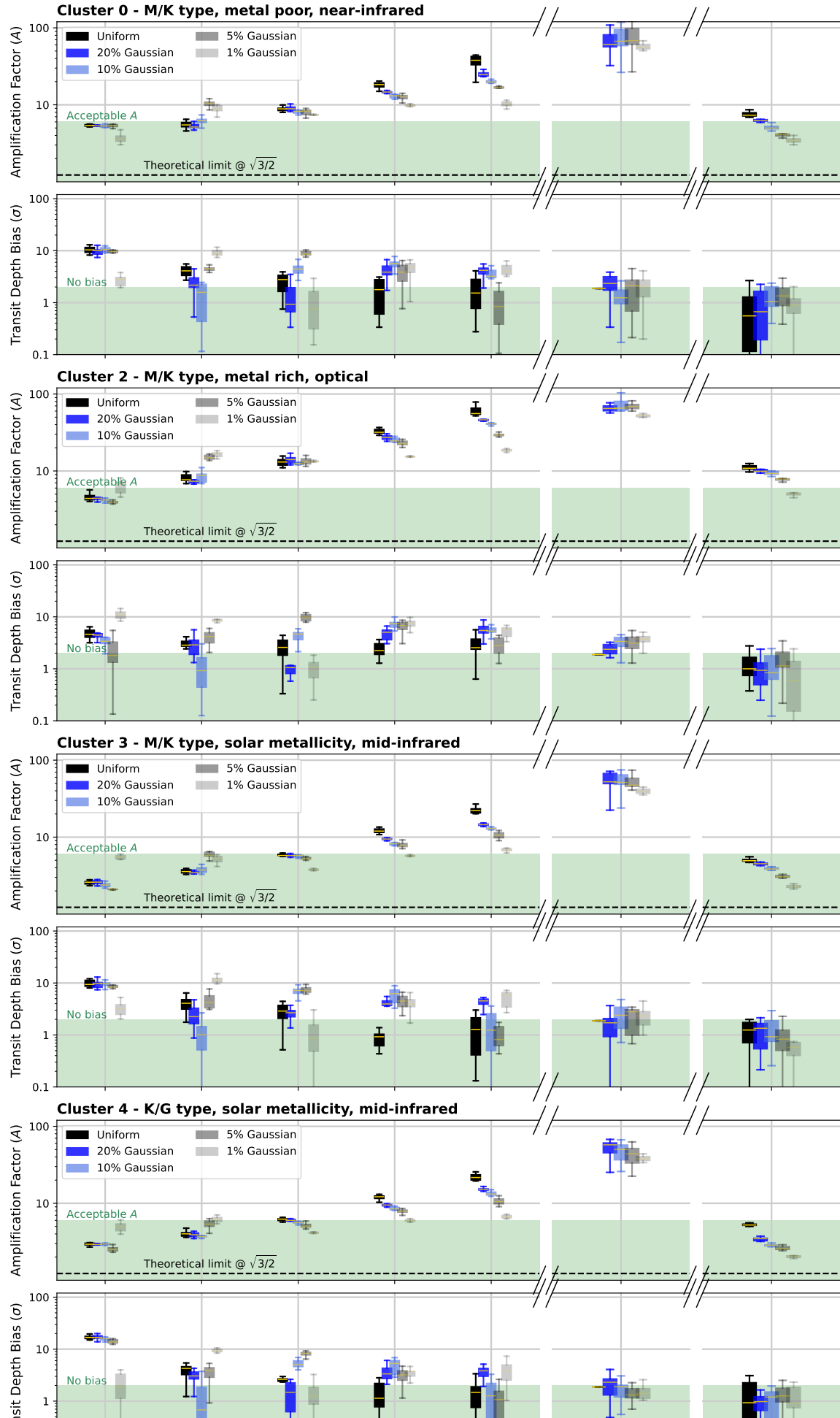
on them affects both the amplification factor and the transit depth bias. To set realistic priors, we estimate the spread in LDCs between the MPS-ATLAS set1 and set2 models (Kostogryz et al. 2022, 2023). The set1 LDC distribution was already derived in Section 4.2; we therefore generate a parallel set2 distribution using identical T_{eff} , $\log g$, and $[M/H]$ grids, and wavelength-filtering steps. The two distributions are compared point-by-point by matching profiles at identical physical parameters, and computing the signed relative difference $(c_{\text{set2}} - c_{\text{set1}}) / |c_{\text{set1}}|$ for each pair. The resulting distributions are summarized as histograms in the top panels of Figure E5. The medians are near zero for all four coefficients ($\leq 1.0\%$), confirming that the two sets are effectively unbiased relative to one another. The inter-model spread, however, grows with coefficient order: the IQR is 5% for c_1 , 15% for c_2 , and 18–19% for c_3 and c_4 . This trend is expected, as higher-order coefficients capture increasingly fine-scale variations in the intensity profile which are, by definition, weaker and harder to constrain. We therefore adopt the IQR of the higher-order coefficients (~ 10 –20%) as the tightest justifiable Gaussian prior width that can be placed on LDCs from current stellar models. We note, however, that the distributions exhibit long, asymmetric tails, and the bottom panels of Figure E5 show that spread grows substantially at extreme stellar parameters—particularly at $[M/H] \lesssim -3$, $T_{\text{eff}} \gtrsim 5500$ K, and short wavelengths. The adopted prior widths should therefore be seen as summary statistics, with inter-model differences potentially shrinking or growing substantially depending on the physical regime considered.

We restrict this comparison to the MPS-ATLAS family, because the two sets share the same underlying atmospheric assumptions⁵ and span identical grids of stellar parameters and wavelengths, simplifying comparison. Other widely used models (*e.g.*, PHOENIX, Husser et al. 2013; Stagger, Magic et al. 2013; Kurucz, Kurucz 1979) are deliberately excluded as their non-overlapping parameter coverage and distinct underlying physics would conflate genuine physical differences with model specification artifacts. Nevertheless, a full sensitivity analysis comparing all available models to photometric measurements, akin to EJ15, remains necessary to fully validate the LDCs derived from this model set.

6.2. Impact of priors and LDL choice

Figure 5 presents our injection–retrieval results in three panels, one per cluster mode used as the injection LDC vector (clusters 1, 3, and 4 in Figure 3; see Sec-

⁵ Differing in their elemental abundances and convection modeling.



tion 4.2). The mock light curves were generated with a fourth-order non-linear LDL (as discussed in Section 4) and injected with 16 ppm of Gaussian white noise. Each panel sweeps over Gaussian priors of relative width⁶ $f \in \{20, 10, 5, 1\}\%$ alongside an uninformative wide uniform prior, and over a broad range of retrieval LDLs: second-, third-, fourth-, fifth-, sixth-, and ninth-order LDLs, as well as fourth-order non-linear LDL. Because the injection and retrieval LDLs generally differ, the prior means cannot simply be set to the injected coefficients. Instead, we fit the profile produced by the injected fourth-order non-linear LDL with whichever retrieval LDL is under consideration and use the resulting LDCs as the Gaussian prior means. The three injection vectors are chosen to span distinct regions of the $(T_{\text{eff}}, \log g, [\text{M}/\text{H}], \lambda)$ -space, allowing us to assess how robustly our findings generalize. All other aspects of the model setup, posterior sampling, and post-processing match Section 3 and are summarized in Table B1.

Five main conclusions emerge. **First, the often-used quadratic (second-order polynomial) LDL is inadequate.** Despite achieving a low amplification factor ($A \sim 3\text{--}6$), it results in substantial transit depth biases for nearly all prior widths tested. The magnitude of this bias scales with the portion of true intensity profile that lies outside the quadratic basis. Concretely, the c_2 and c_4 terms of the injected fourth-order non-linear LDL are exactly representable as quadratic combinations of $(1 - \mu)$, but the half-integer-power terms $c_1(1 - \mu^{1/2})$ and $c_3(1 - \mu^{3/2})$ are not, and their residuals are precisely what the transit depth must absorb. This explains why bias does not simply track the overall magnitude of LDCs. Although cluster 1 has the largest $|c_i|$ values of the three modes, its c_1 and c_3 share the same sign and similar amplitude, so their residuals partially cancel. The same is true for cluster 4. Conversely, in cluster 3, $c_1 \approx 0$ removes the cancellation entirely, leaving the full c_3 contribution to drive non-quadratic behavior and bias the transit depth. These expectations match what we observe in Figure 5: the quadratic LDL yields biases of $\sim 5\text{--}6\sigma$ for clusters 1 and 4, but $\sim 10\sigma$ for cluster 3. The same logic governs how rapidly bias improves with higher-order LDLs ($\sim 10\sigma \rightarrow \sim 1.5\sigma$ in cluster 3, $\sim 6\sigma \rightarrow \sim 3\sigma$ in cluster 1; $\sim 5\sigma \rightarrow \sim 2.5\sigma$ in cluster 4): the more residual is left for the depth to absorb under a low-order LDL, the more there is to recover when flexibility is added. We conclude that reducing the number of LDCs is not a viable route to mitigating the TDPP:

if the chosen LDL lacks the expressivity to capture the true profile, large biases are unavoidable. Importantly, the bias incurred from misspecifying the intensity profile is *absorbed into D , not into the fitted quadratic LDCs themselves*, which is precisely why such issues cannot be diagnosed by checking LDC values against model predictions.

Second, the inadequacy of the quadratic law is strongly wavelength-dependent, reflecting the chromatic structure of the LDC space itself. The cluster-to-cluster variation described above is not coincidental: the LDC distribution is intrinsically wavelength-dependent (see Figure C1). In the optical (such as cluster 3 at $\lambda = 0.88 \mu\text{m}$), c_2 is large and positive while c_3 is large and negative, so the non-quadratic residual is maximized. At longer wavelengths (such as the global mode at $\lambda = 3.21 \mu\text{m}$), c_2 is mildly negative and c_3 is small, reducing the residual and hence the bias. The cluster-to-cluster variation in quadratic bias therefore reflects, in part, an underlying wavelength dependence in the LDC structure itself. Such a dependence, owing to its complexity, wealth of information, and inherent relation to T_{eff} , $\log g$, and $[\text{M}/\text{H}]$ will be explored in future works.

Third, polynomials of third-order and higher result in more reliable retrievals, generally producing biases $\leq 2\sigma$ when paired with priors at the level currently achievable from stellar models. The third-order polynomial LDL emerges as a practical “sweet spot”: flexible enough to approximate the true intensity profile without introducing excessive degeneracy, yet not so high dimensional that the amplification factor becomes unmanageable. At fourth-order and beyond, the added degrees of freedom inflate A rapidly without any compensating gain in accuracy: for cluster 4 under a uniform prior, A reaches ~ 10 at fourth-order, ~ 50 at sixth-order, and exceeds 100 at ninth-order—placing the corresponding box plot above the plotted range. This behavior is the expected consequence of over-parametrization: with nine free LDCs and no informative priors, the higher-order polynomial LDCs are essentially unconstrained by the data, and the resulting LDC- D degeneracies inflate D ’s distribution.

Fourth, the fourth-order non-linear LDL performs best. Functionally matching the injection model, it results in the lowest transit depth bias and amplification factor. Tightening priors from uniform to 10–20% width Gaussian reduces A by a factor of 1.2–1.8, and pushing priors further to 1% width—potentially achievable with future stellar models—yields a 2.3–2.5 $\times$ decrease. Such gains in transit depth precision translate to requiring 3–6 $\times$ fewer transits to achieve the same

⁶ Relative meaning that for a LDC with true value 0.1, a width of 20% translates to a Gaussian prior with mean 0.1 and standard deviation $\sigma = 0.2 \times 0.1 = 0.02$.

level of scientific return (*e.g.*, placing constraints on atmospheric properties from transmission spectra).

Finally, the trends with prior strength reveal a tension. Tighter priors consistently reduce A , as expected, since narrower priors constrain both the LDC and transit depth posteriors. The corresponding effect on bias is less clear. For polynomial retrievals, biases broadly grow as priors tighten, due to the prior-centering procedure described above. Because polynomial LDLs cannot perfectly reproduce the injected non-linear profile, especially at low orders, the LDC prior means are intrinsically shifted away from optimum. Tightening around them confines the chains to biased regions of parameter space and other parameters such as the transit depth shift to compensate, inflating their biases. However, as we observe it here, the trend with prior width is not monotonic. This likely reflects two effects: the limited statistical power of 10 noise realizations per configuration, and the fact that prior-induced biases on different LDCs do not necessarily point in the same direction, so their net effect on D can vary non-trivially across different injection–retrievals. The fourth-order non-linear LDL is immune to this tension since it functionally matches the injection model, meaning tighter priors reduce A without inflating bias.

Taken together, these results suggest that observers should fit modern light curves with, both a third-order polynomial and fourth-order non-linear LDL. Both prescriptions provide an optimal compromise between A and bias, while also serving as useful cross-checks: their distinct functional forms make them sensitive to different aspects of the stellar intensity profile, enabling a more complete characterization of the underlying stellar properties. As a community, we should therefore encourage the use of multiple LDLs when fitting light curves and recommend reporting results from each prescription to highlight whether transit depth measurements differ or remain consistent across models. Particular care should also be paid to the wavelength dependence of limb-darkening parametrization (exemplified in Figure C1), since inadequate choices can introduce spurious features in transmission spectra that mimic genuine atmospheric signals (*e.g.*, Keers et al. 2024).

7. CONCLUSION

We have examined why the precision of exoplanet transit depth measurements (D) often falls short of theoretical expectations, a shortfall we define as the Transit Depth Precision Problem (TDPP). Through an injection–retrieval framework, we show that the uncertainty on D is systematically amplified by a factor of $A \gg \sqrt{3}$ while biases arise from inadequate limb-

darkening parameterizations. Crucially, these effects are not independent: choices that reduce A (tightening LDC priors or lowering the LDL order) often inflate the bias on D , and vice versa. Since bias is inaccessible in real analyses but transit depth uncertainty is, we recommend that future studies report A in each wavelength bin as a sanity check; $A \gg 6$ would suggest that tighter constraints are accessible, while $A \ll 6$ implies artificially tight constraints and potentially biased measurements.

Our perturbation–sensitivity analyses link the TDPP to correlations between D and the LDCs. These correlations broaden D ’s posterior, while any portion of the true intensity profile that the chosen LDL cannot capture is absorbed into D itself, biasing its central value. The size of each effect depends on the chosen LDL, the number of free LDCs, and the priors imposed on them.

Near-term mitigation. Our results disfavor the quadratic LDL, which yields biases of $\sim 5\text{--}10\sigma$ despite its low A of $\sim 3\text{--}6$. We instead favor the third-order polynomial and fourth-order non-linear LDLs which optimally balance A and bias. With model-informed LDC priors at the 10–20% level, both yield $A \sim 5\text{--}6$ and bias $\leq 2\sigma$. Tested on representative LDC vectors drawn from a hierarchical clustering of $> 500,000$ intensity profiles, this behavior holds across the (T_{eff} , $\log g$, $[M/H]$, λ)-space probed, though caution is warranted at short-wavelengths and for low metallicity or low effective temperature stars, where the LDC structure is most non-quadratic and biases are largest.

Long-term solutions. While the dual-LDL strategy presented is a reliable near-term workaround, it does not eliminate the TDPP at its root. Genuine resolution requires two complementary improvements. First, next-generation, high-fidelity stellar models are needed to enable precise and accurate priors on LDCs—particularly in the regimes where intensity profiles are most complex. Second, transit fitting tools must adopt novel limb-darkening parameterizations that reduce D -LDC correlations (*e.g.*, more orthogonal bases as in Maxted 2018, 2023) and exploit the wavelength-dependence of intensity profiles in multi-wavelength transit fitting.

In conclusion, the limiting factor in transit depth precision and accuracy is not instrumental and future advances in stellar models and transit fitting tools could jointly reduce A and bias—enabling the same science with up to $\sim 5\times$ fewer transits, and fully exploiting the information content of high-quality data from JWST and future observatories. In the meantime, a dual-LDL strategy is the most effective way to mitigate the bias incurred by the quadratic law.

ACKNOWLEDGMENTS

We thank the reviewer for a careful and constructive review, which has substantially strengthened the manuscript. This material is based upon work supported by the European Research Council (ERC) Synergy Grant under the European Union’s Horizon 2020 research and innovation program (grant No. 101118581—project REVEAL). This material is based upon work supported by the National Aeronautics and Space Administration under Agreement No. 80NSSC21K0593 for the program “Alien Earths”. The results reported herein benefited from collaborations and/or information exchange within NASA’s Nexus for Exoplanet System Science (NExSS) research coordination network sponsored by NASA’s Science Mission Directorate. SJM acknowledges support from the Massachusetts Institute of Technology through the Praecis Presidential Fellowship and Desmond Fellowship.

Software: `arviz`⁷(v0.20.0, Martin et al. 2026), `astropy`⁸(v7.2.0, Astropy Collaboration et al. 2013, 2018, 2022), `corner`⁹(v2.2.2, Foreman-Mackey 2016), `emcee-jax`¹⁰ (v0.1.dev38), `ExoTiC-LD`¹¹(v3.2.0, Grant & Wakeford 2024), `jax`¹²(v0.9.1, Bradbury et al. 2018), `jaxoplanet`¹³(v0.1.0, Hattori et al. 2024), `lmfit`¹⁴(v1.3.3, Newville et al. 2016), `Matplotlib`¹⁵(v3.10.8, Hunter 2007), `numpy`¹⁶(v2.4.3, Harris et al. 2020), `numpyro`¹⁷(v0.19.0, Phan et al. 2019), `pandas`¹⁸(v2.2.3, The pandas development Team 2025), `scikit-learn`¹⁹(v1.6.1, Pedregosa et al. 2011), `scipy`²⁰(v1.17.0, Virtanen et al. 2020), `squishyplanet`²¹(v0.3.1, Cassese et al. 2024), `tqdm`²²(v4.67.1, da Costa-Luis 2019),

REFERENCES

- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, *A&A*, 558, A33, doi: [10.1051/0004-6361/201322068](https://doi.org/10.1051/0004-6361/201322068)
- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, *AJ*, 156, 123, doi: [10.3847/1538-3881/aabc4f](https://doi.org/10.3847/1538-3881/aabc4f)
- Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, *apj*, 935, 167, doi: [10.3847/1538-4357/ac7c74](https://doi.org/10.3847/1538-4357/ac7c74)
- Benneke, B., & Seager, S. 2012, *ApJ*, 753, 100, doi: [10.1088/0004-637X/753/2/100](https://doi.org/10.1088/0004-637X/753/2/100)
- Benneke, B., Roy, P.-A., Coulombe, L.-P., et al. 2024, *arXiv e-prints*, arXiv:2403.03325, doi: [10.48550/arXiv.2403.03325](https://doi.org/10.48550/arXiv.2403.03325)
- Berardo, D., & de Wit, J. 2022, *ApJ*, 935, 178, doi: [10.3847/1538-4357/ac82b2](https://doi.org/10.3847/1538-4357/ac82b2)
- Bourque, M., Espinoza, N., Filippazzo, J., et al. 2021, The Exoplanet Characterization Toolkit (ExoCTK), 1.0.0, Zenodo, doi: [10.5281/zenodo.4556063](https://doi.org/10.5281/zenodo.4556063)
- Bradbury, J., Frostig, R., Hawkins, P., et al. 2018, JAX: composable transformations of Python+NumPy programs, 0.3.13. <http://github.com/jax-ml/jax>
- Carter, J. A., Winn, J. N., Holman, M. J., et al. 2011, *ApJ*, 730, 82, doi: [10.1088/0004-637X/730/2/82](https://doi.org/10.1088/0004-637X/730/2/82)
- Carter, J. A., Yee, J. C., Eastman, J., Gaudi, B. S., & Winn, J. N. 2008, *ApJ*, 689, 499, doi: [10.1086/592321](https://doi.org/10.1086/592321)
- Cassese, B., Vega, J., Lu, T., et al. 2024, *The Journal of Open Source Software*, 9, 6972, doi: [10.21105/joss.06972](https://doi.org/10.21105/joss.06972)
- Charbonneau, D., Brown, T. M., Latham, D. W., & Mayor, M. 2000, *ApJL*, 529, L45, doi: [10.1086/312457](https://doi.org/10.1086/312457)
- Claret, A. 1998, *A&A*, 335, 647
- . 2000, *A&A*, 363, 1081
- Claret, A., & Bloemen, S. 2011, *A&A*, 529, A75, doi: [10.1051/0004-6361/201116451](https://doi.org/10.1051/0004-6361/201116451)
- Csizmadia, S., Pasternacki, T., Dreyer, C., et al. 2013, *A&A*, 549, A9, doi: [10.1051/0004-6361/201219888](https://doi.org/10.1051/0004-6361/201219888)
- da Costa-Luis, C. 2019, *The Journal of Open Source Software*, 4, 1277, doi: [10.21105/joss.01277](https://doi.org/10.21105/joss.01277)
- de Wit, J., Gillon, M., Demory, B. O., & Seager, S. 2012, *A&A*, 548, A128, doi: [10.1051/0004-6361/201219060](https://doi.org/10.1051/0004-6361/201219060)
- de Wit, J., & Seager, S. 2013, *Science*, 342, 1473, doi: [10.1126/science.1245450](https://doi.org/10.1126/science.1245450)
- Deming, D., Seager, S., Winn, J., et al. 2009, *PASP*, 121, 952, doi: [10.1086/605913](https://doi.org/10.1086/605913)
- Dumusque, X., Santos, N. C., Udry, S., Lovis, C., & Bonfils, X. 2011a, *A&A*, 527, A82, doi: [10.1051/0004-6361/201015877](https://doi.org/10.1051/0004-6361/201015877)

⁷ <https://www.arviz.org/en/latest/>

⁸ <https://www.astropy.org>

⁹ <https://corner.readthedocs.io/en/latest/>

¹⁰ <https://github.com/SamMerc/emcee-jax>

¹¹ <https://exotic-ld.readthedocs.io/en/latest/>

¹² <https://docs.jax.dev/en/latest/>

¹³ <https://jax.exoplanet.codes/en/latest/>

¹⁴ <https://lmfit.github.io/lmfit-py/>

¹⁵ <https://matplotlib.org>

¹⁶ <https://numpy.org>

¹⁷ <https://num.pyro.ai/en/stable/>

¹⁸ <https://pandas.pydata.org>

¹⁹ <https://scikit-learn.org/stable/>

²⁰ <https://scipy.org>

²¹ <https://squishyplanet.readthedocs.io/en/latest/>

²² <https://tqdm.github.io>

- Dumusque, X., Udry, S., Lovis, C., Santos, N. C., & Monteiro, M. J. P. F. G. 2011b, *A&A*, 525, A140, doi: [10.1051/0004-6361/201014097](https://doi.org/10.1051/0004-6361/201014097)
- Dyrek, A., Min, M., Decin, L., et al. 2024, *Nature*, 625, 51, doi: [10.1038/s41586-023-06849-0](https://doi.org/10.1038/s41586-023-06849-0)
- Espinoza, N., & Jordán, A. 2015, *MNRAS*, 450, 1879, doi: [10.1093/mnras/stv744](https://doi.org/10.1093/mnras/stv744)
- . 2016, *MNRAS*, 457, 3573, doi: [10.1093/mnras/stw224](https://doi.org/10.1093/mnras/stw224)
- Everitt, B. S., Landau, S., Leese, M., & Stahl, D. 2011, *Cluster Analysis*, 5th edn. (Wiley), doi: [10.1002/9780470977811](https://doi.org/10.1002/9780470977811)
- Foreman-Mackey, D. 2016, *The Journal of Open Source Software*, 1, 24, doi: [10.21105/joss.00024](https://doi.org/10.21105/joss.00024)
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, *PASP*, 125, 306, doi: [10.1086/670067](https://doi.org/10.1086/670067)
- Giménez, A. 2006, *A&A*, 450, 1231, doi: [10.1051/0004-6361:20054445](https://doi.org/10.1051/0004-6361:20054445)
- Grant, D., & Wakeford, H. R. 2024, *Journal of Open Source Software*, 9, 6816, doi: [10.21105/joss.06816](https://doi.org/10.21105/joss.06816)
- Greene, T. P., Line, M. R., Montero, C., et al. 2016, *ApJ*, 817, 17, doi: [10.3847/0004-637X/817/1/17](https://doi.org/10.3847/0004-637X/817/1/17)
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Nature*, 585, 357, doi: [10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- Hattori, S., Garcia, L., Murray, C., et al. 2024, *exoplanet-dev/jaxoplanet: Astronomical time series analysis with JAX, v0.0.2*, Zenodo, doi: [10.5281/zenodo.10736936](https://doi.org/10.5281/zenodo.10736936)
- Henry, G. W., Marcy, G. W., Butler, R. P., & Vogt, S. S. 2000, *ApJL*, 529, L41, doi: [10.1086/312458](https://doi.org/10.1086/312458)
- Howarth, I. D. 2011, *MNRAS*, 418, 1165, doi: [10.1111/j.1365-2966.2011.19568.x](https://doi.org/10.1111/j.1365-2966.2011.19568.x)
- Hunter, J. D. 2007, *Computing in Science & Engineering*, 9, 90, doi: [10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)
- Husser, T. O., Wende-von Berg, S., Dreizler, S., et al. 2013, *A&A*, 553, A6, doi: [10.1051/0004-6361/201219058](https://doi.org/10.1051/0004-6361/201219058)
- Keers, R. E., Shapiro, A. I., Kostogryz, N. M., et al. 2024, *ApJL*, 977, L7, doi: [10.3847/2041-8213/ad8b51](https://doi.org/10.3847/2041-8213/ad8b51)
- Kipping, D. M. 2013, *MNRAS*, 435, 2152, doi: [10.1093/mnras/stt1435](https://doi.org/10.1093/mnras/stt1435)
- . 2016, *MNRAS*, 455, 1680, doi: [10.1093/mnras/stv2379](https://doi.org/10.1093/mnras/stv2379)
- Kostogryz, N., Shapiro, A., Witzke, V., et al. 2023, *Research Notes of the AAS*, 7, 39
- Kostogryz, N., Witzke, V., Shapiro, A., et al. 2022, *Astronomy & Astrophysics*, 666, A60
- Kurucz, R. L. 1979, *ApJS*, 40, 1, doi: [10.1086/190589](https://doi.org/10.1086/190589)
- Liu, Q., Zhu, W., Zhou, Y., et al. 2025, *AJ*, 169, 79, doi: [10.3847/1538-3881/ad9b8c](https://doi.org/10.3847/1538-3881/ad9b8c)
- Lovis, C., Dumusque, X., Santos, N. C., et al. 2011, *arXiv e-prints*, arXiv:1107.5325, doi: [10.48550/arXiv.1107.5325](https://doi.org/10.48550/arXiv.1107.5325)
- Magic, Z., Collet, R., Asplund, M., et al. 2013, *A&A*, 557, A26, doi: [10.1051/0004-6361/201321274](https://doi.org/10.1051/0004-6361/201321274)
- Manduca, A., Bell, R. A., & Gustafsson, B. 1977, *A&A*, 61, 809
- Martin, O. A., Abril-Pla, O., Deklerk, J., et al. 2026, *Journal of Open Source Software*, 11, 9889, doi: [10.21105/joss.09889](https://doi.org/10.21105/joss.09889)
- Maxted, P. F. L. 2018, *A&A*, 616, A39, doi: [10.1051/0004-6361/201832944](https://doi.org/10.1051/0004-6361/201832944)
- Maxted, P. F. L. 2023, *MNRAS*, 519, 3723, doi: [10.1093/mnras/stac3741](https://doi.org/10.1093/mnras/stac3741)
- Meunier, N., Mignon, L., & Lagrange, A. M. 2017, *A&A*, 607, A124, doi: [10.1051/0004-6361/201731017](https://doi.org/10.1051/0004-6361/201731017)
- Milne, E. A. 1921, *MNRAS*, 81, 361, doi: [10.1093/mnras/81.5.361](https://doi.org/10.1093/mnras/81.5.361)
- Morello, G., Tsiaras, A., Howarth, I. D., & Homeier, D. 2017, *AJ*, 154, 111, doi: [10.3847/1538-3881/aa8405](https://doi.org/10.3847/1538-3881/aa8405)
- Neilson, H. R., & Lester, J. B. 2013a, *A&A*, 554, A98, doi: [10.1051/0004-6361/201321502](https://doi.org/10.1051/0004-6361/201321502)
- . 2013b, *A&A*, 556, A86, doi: [10.1051/0004-6361/201321888](https://doi.org/10.1051/0004-6361/201321888)
- Newville, M., Stensitzki, T., Allen, D. B., et al. 2016, *Lmfit: Non-Linear Least-Square Minimization and Curve-Fitting for Python*, *Astrophysics Source Code Library*, record ascl:1606.014. <http://ascl.net/1606.014>
- Niraula, P., de Wit, J., Gordon, I. E., Hargreaves, R. J., & Sousa-Silva, C. 2023, *ApJL*, 950, L17, doi: [10.3847/2041-8213/acd6f8](https://doi.org/10.3847/2041-8213/acd6f8)
- Niraula, P., de Wit, J., Gordon, I. E., et al. 2022, *Nature Astronomy*, 6, 1287, doi: [10.1038/s41550-022-01773-1](https://doi.org/10.1038/s41550-022-01773-1)
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, *Journal of Machine Learning Research*, 12, 2825
- Phan, D., Pradhan, N., & Jankowiak, M. 2019, *arXiv e-prints*, arXiv:1912.11554, doi: [10.48550/arXiv.1912.11554](https://doi.org/10.48550/arXiv.1912.11554)
- Rabus, M., Alonso, R., Belmonte, J. A., et al. 2009, *A&A*, 494, 391, doi: [10.1051/0004-6361:200811110](https://doi.org/10.1051/0004-6361:200811110)
- Rackham, B., Espinoza, N., Apai, D., et al. 2017, *ApJ*, 834, 151, doi: [10.3847/1538-4357/aa4f6c](https://doi.org/10.3847/1538-4357/aa4f6c)
- Rackham, B. V., Apai, D., & Giampapa, M. S. 2018, *ApJ*, 853, 122, doi: [10.3847/1538-4357/aaa08c](https://doi.org/10.3847/1538-4357/aaa08c)
- . 2019, *AJ*, 157, 96, doi: [10.3847/1538-3881/aaf892](https://doi.org/10.3847/1538-3881/aaf892)
- Rackham, B. V., Espinoza, N., Berdyugina, S. V., et al. 2023, *RAS Techniques and Instruments*, 2, 148, doi: [10.1093/rasti/rzad009](https://doi.org/10.1093/rasti/rzad009)
- Saba, A., Thompson, A., Yip, K. H., et al. 2025, *ApJS*, 276, 70, doi: [10.3847/1538-4365/ad8c3c](https://doi.org/10.3847/1538-4365/ad8c3c)
- Seager, S., & Mallén-Ornelas, G. 2003, *ApJ*, 585, 1038, doi: [10.1086/346105](https://doi.org/10.1086/346105)

- 947 Seager, S., & Sasselov, D. D. 2000, *ApJ*, 537, 916,
948 doi: [10.1086/309088](https://doi.org/10.1086/309088)
- 949 Sing, D. K. 2010, *A&A*, 510, A21,
950 doi: [10.1051/0004-6361/200913675](https://doi.org/10.1051/0004-6361/200913675)
- 951 Southworth, J. 2008, *MNRAS*, 386, 1644,
952 doi: [10.1111/j.1365-2966.2008.13145.x](https://doi.org/10.1111/j.1365-2966.2008.13145.x)
- 953 The pandas development Team. 2025, *pandas-dev/pandas*:
954 *Pandas*, v2.3.1, Zenodo, doi: [10.5281/zenodo.3509134](https://doi.org/10.5281/zenodo.3509134)
- 955 Tsai, S.-M., Innes, H., Lichtenberg, T., et al. 2021, *ApJL*,
956 922, L27, doi: [10.3847/2041-8213/ac399a](https://doi.org/10.3847/2041-8213/ac399a)
- 957 Tsai, S.-M., Lee, E. K. H., Powell, D., et al. 2023, *Nature*,
958 617, 483, doi: [10.1038/s41586-023-05902-2](https://doi.org/10.1038/s41586-023-05902-2)
- 959 Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020,
960 *Nature Methods*, 17, 261, doi: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2)
- 961 Ward, J. H. 1963, *Journal of the American Statistical*
962 *Association*, 58, 236
- 963 Welbanks, L., Bell, T. J., Beatty, T. G., et al. 2024, *Nature*,
964 630, 836, doi: [10.1038/s41586-024-07514-w](https://doi.org/10.1038/s41586-024-07514-w)
- 965 Xue, Q., Bean, J. L., Zhang, M., et al. 2024, *ApJL*, 963, L5,
966 doi: [10.3847/2041-8213/ad2682](https://doi.org/10.3847/2041-8213/ad2682)

APPENDIX

A. DERIVATION OF THE THEORETICAL AMPLIFICATION FACTOR LIMIT

Consider a transit light curve composed of three equal duration segments: a pre-transit baseline, an in-transit segment, and a post-transit baseline, each containing N points. Let σ_{OOT} denote the typical per-point photometric uncertainty, assumed identical across all segments and uncorrelated. The transit depth D is measured as the difference between the out-of-transit mean flux and the in-transit mean flux. These two quantities are estimated independently from disjoint sets of points, so their uncertainties combine in quadrature. The uncertainty on the in-transit mean is $\sigma_{\text{OOT}}/\sqrt{N}$, and the uncertainty on the out-of-transit mean, estimated from $2N$ baseline points, is $\sigma_{\text{OOT}}/\sqrt{2N}$. The uncertainty on their difference is therefore

$$\sigma_D = \sigma_{\text{OOT}} \sqrt{\frac{1}{N} + \frac{1}{2N}} = \sigma_{\text{OOT}} \sqrt{\frac{3}{2N}}. \quad (\text{A1})$$

With $\sigma_{\text{ref}} \equiv \sigma_{\text{OOT}}/\sqrt{N}$, the true theoretical floor is therefore $A_{\text{theory}} = \sigma_D/\sigma_{\text{ref}} = \sqrt{3/2}$. While we reference the aforementioned bound throughout the manuscript, it is important to underline its dependence on the choice of baseline. Indeed, while equal duration pre-, post-, and in-transit segments are ideal (and, in fact, recommended by telescope guidelines XXX) acquiring such temporal coverage is not always possible and a large proportion of single-transit observations instead use a 1:1 baseline, when only pre- or post-transit data are available. In this case (which is also recommended by other telescope guidelines XXX), the bound increases to $A_{\text{theory}} = \sqrt{2}$. Finally, we note that one could further increase the bound σ_D by erroneously combining the two correlated difference estimators (pre-minus-in transit and post-minus-in transit), yielding $A_{\text{theory}} = 2$. While these differences highlight the baseline-dependence of the theoretical limit, no matter the value selected the observed amplification factor values (*e.g.*, Figure 1, Figure 5) represents a dramatic departure from any of these theoretical expectations, and none of our quantitative results reported throughout the paper are relative to this theoretical limit, which serves a purely referential purpose for users to remind themselves of what the true limit is. Given that the results of our study are not affected by the choice of limit, we default to using the 2:1 baseline value of $A_{\text{theory}} = \sqrt{3/2}$.

B. SYSTEM AND INJECTION-RETRIEVAL PARAMETERS

Table B1 provides the parameters of the injection-retrieval framework used throughout the text.

Table B1. Injection-retrieval summary

Parameter	Value						
<i>Injection - System parameters</i>							
$R_p/R_\star$	0.1						
i (degrees)	90						
P (days)	1						
$a/R_\star$	~ 4.21 (Kepler's third law)						
e	0						
ω (degrees)	0						
T_0 (days)	0						
$R_\star$ ($R_\odot$)	1						
$M_\star$ ($M_\odot$)	1						
<i>Injection - LDCs</i>							
	Section 3 (illustrative)	Section 5 (Global mode)	(Cluster 0)	(Cluster 2)	Section 6 (Cluster 3)	(Cluster 4)	(Cluster 7)
c_1	0.1	0.6245	0.5597	0.7617	0.6666	0.5172	0.6383
c_2	0.2	−0.1898	−0.1310	−0.8502	−0.8767	−0.1913	−0.0511
c_3	0.4	0.1473	0.4556	1.4229	0.8343	0.0819	−0.2722
c_4	0.3	−0.0634	−0.2398	−0.4899	−0.2968	−0.0316	0.1455
<i>Corresponding stellar parameters</i>							
	—	Global mode	Cluster 0	Cluster 2	Cluster 3	Cluster 4	Cluster 7
T_{eff} (K)	—	8389	4111	4111	4111	5333	5944
$\log g$	—	4.33	3.67	4.78	3.89	3.00	4.33
[M/H]	—	0.06	−0.67	0.78	0.06	0.06	0.06
λ (μm)	—	0.86	1.30	0.71	5.27	3.11	2.02
<i>Retrieval</i>							
Parameter	Prior				Walker init.		
$R_P/R_\star$	$\mathcal{U}(0, 1)$				$\mathcal{U}(0.07, 0.15)$		
i (degrees)	$\mathcal{U}(70, 110)$				$\mathcal{U}(88, 92)$		
P (days)	$\mathcal{N}(1, 5 \times 10^{-4})$				$\mathcal{U}(0.9995, 1.0005)$		
$a/R_\star$	$\mathcal{U}(0, 50)$				$\mathcal{U}(a/R_\star - 2, a/R_\star + 2)$		
$\sqrt{e} \cos \omega$	$\mathcal{U}(-1, 1)$				$\mathcal{U}(-0.2, 0.2)$		
$\sqrt{e} \sin \omega$	$\mathcal{U}(-1, 1)$				$\mathcal{U}(-0.2, 0.2)$		
c_i (uniform)	$\mathcal{U}(-100, 100)$				$\mathcal{U}(c_i^{\text{true}} - 0.5, c_i^{\text{true}} + 0.5)$		
c_i (Gaussian)	$\mathcal{N}(c_i^{\text{true}}, f \cdot c_i^{\text{true}})$				$\mathcal{U}(c_i^{\text{true}} - 0.5, c_i^{\text{true}} + 0.5)$		
T_0 (days)	Fixed				—		

NOTE—System parameters describe the hypothetical transit system used throughout the paper. Three sets of injection LDCs are reported, each used in a different section: (1) the illustrative set in Section 3 was chosen to systematically vary the order of the injected LDL (the n^{th} -order law uses the first n values, *e.g.*, the second-order polynomial uses $[0.1, 0.2]$); (2) the global mode in Section 5 is the LDC vector lying closest to its cluster centroid across all nine clusters identified in Section 4.2, representing the most typical regime of the physical LDC space; (3) the three cluster modes in Section 6 sample distinct regions of LDC space to assess how our conclusions generalize across stellar conditions. The stellar parameters reported correspond to the (T_{eff} , $\log g$, [M/H], λ) of the intensity profile from which each mode's LDC vector was derived. Eccentricity and argument of periastron are fixed to circular values during injection but included as free retrieval parameters to characterize their effect on the posterior. The Gaussian LDC priors used in Section 6 are centered on the injected truth with fractional width $f \in \{20, 10, 5, 1\}\%$ as a proxy for prior strength.

C. LIMB-DARKENING DISTRIBUTION AND CLUSTERING

Figure C1 and Figure C2 shows the full distribution of fourth-order non-linear LDCs across T_{eff} , $\log g$, $[M/H]$, and wavelength, while Figure C3 shows the same LDC space partitioned by hierarchical cluster membership.

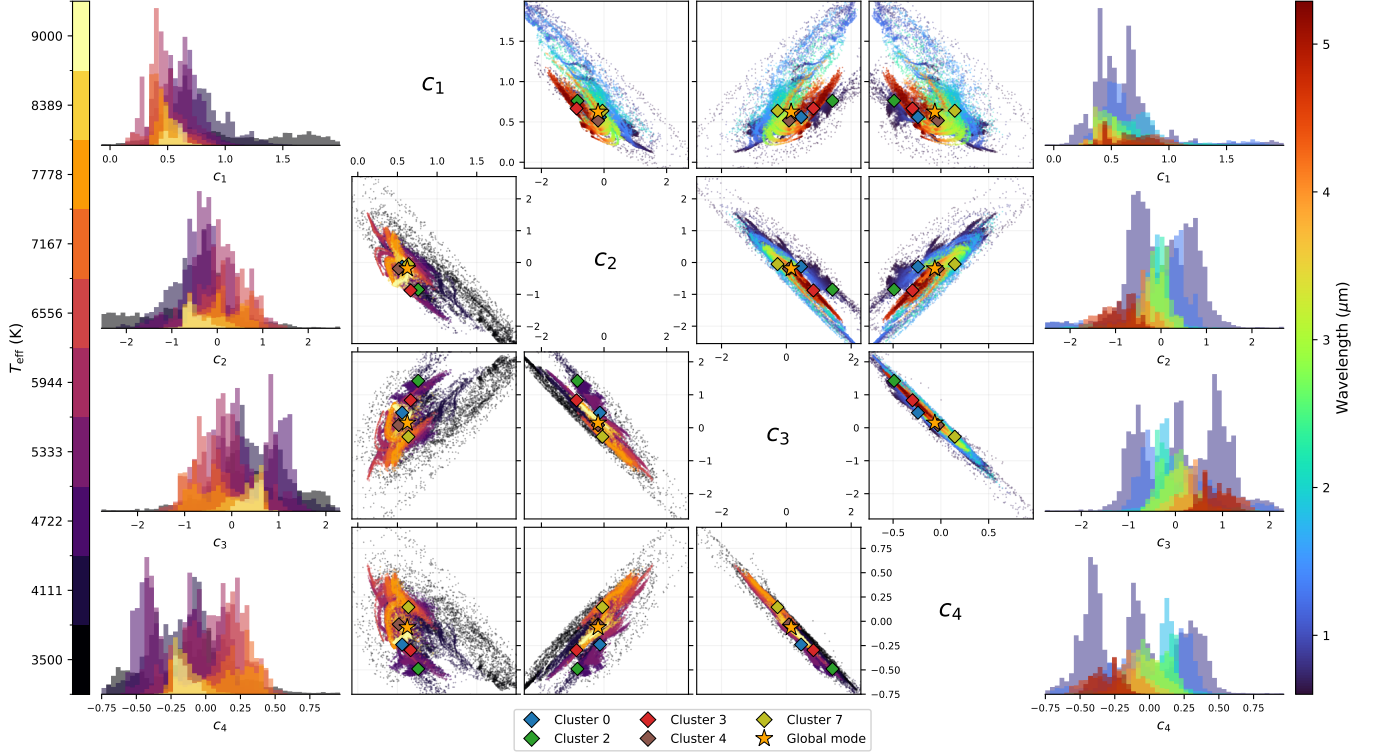


Figure C1. Distribution of fourth-order non-linear limb-darkening coefficients from the MPS-ATLAS set1 grid. Double corner plot of the four LDCs (c_1 , c_2 , c_3 , c_4) computed over $T_{\text{eff}} \in [3500, 9000]$ K, $\log g \in [3.0, 5.0]$, $[M/H] \in [-5.0, 1.5]$, and JWST NIRSpec PRISM wavelengths. The *lower-left* and *upper-right* triangles show LDCs colored by T_{eff} and wavelength, respectively. The star and diamonds mark the location of the global and cluster modes used as injection LDCs throughout the manuscript. Diamond colors reflect the cluster colors used in Figure C3. The star and diamonds mark the location of the global and cluster modes used as injection LDCs throughout the manuscript. The global mode corresponds to a near-solar metallicity K/M-type star at mid-infrared wavelengths, while clusters 1, 3, and 4 respectively probe a metal-poor K/M-type in the near-infrared, a moderately metal-poor G-type in the optical, and a metal-rich K-type in the optical. *Left column*: 1D marginal histograms, grouped by the T_{eff} at which the LDCs were computed. *Right column*: Same as the left column, but with grouping done according to wavelength. An animated version cycling through all four physical dimensions (T_{eff} , $\log g$, $[M/H]$, and wavelength) is available on GitHub.

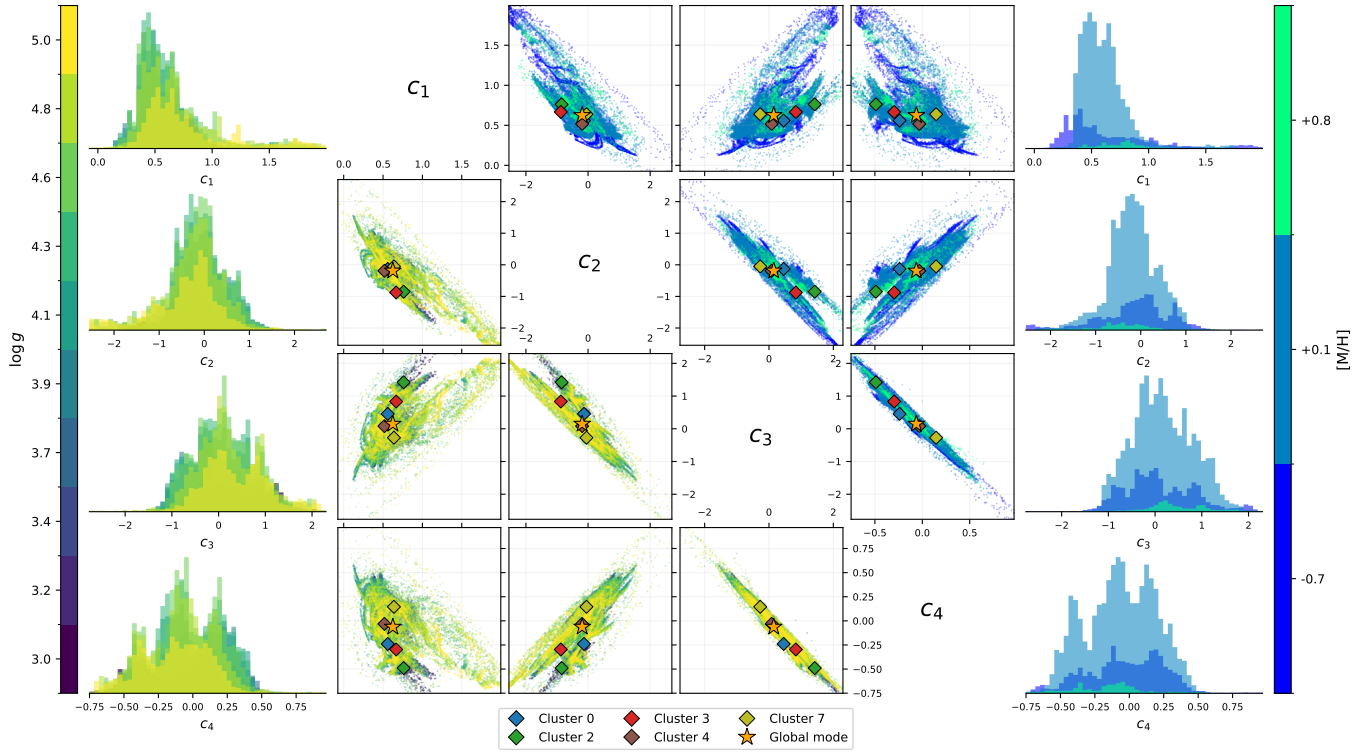


Figure C2. Distribution of fourth-order non-linear limb-darkening coefficients from the MPS-ATLAS set1 grid.

Same as Figure C1 except coloring LDCs by $\log g$ and $[M/H]$.

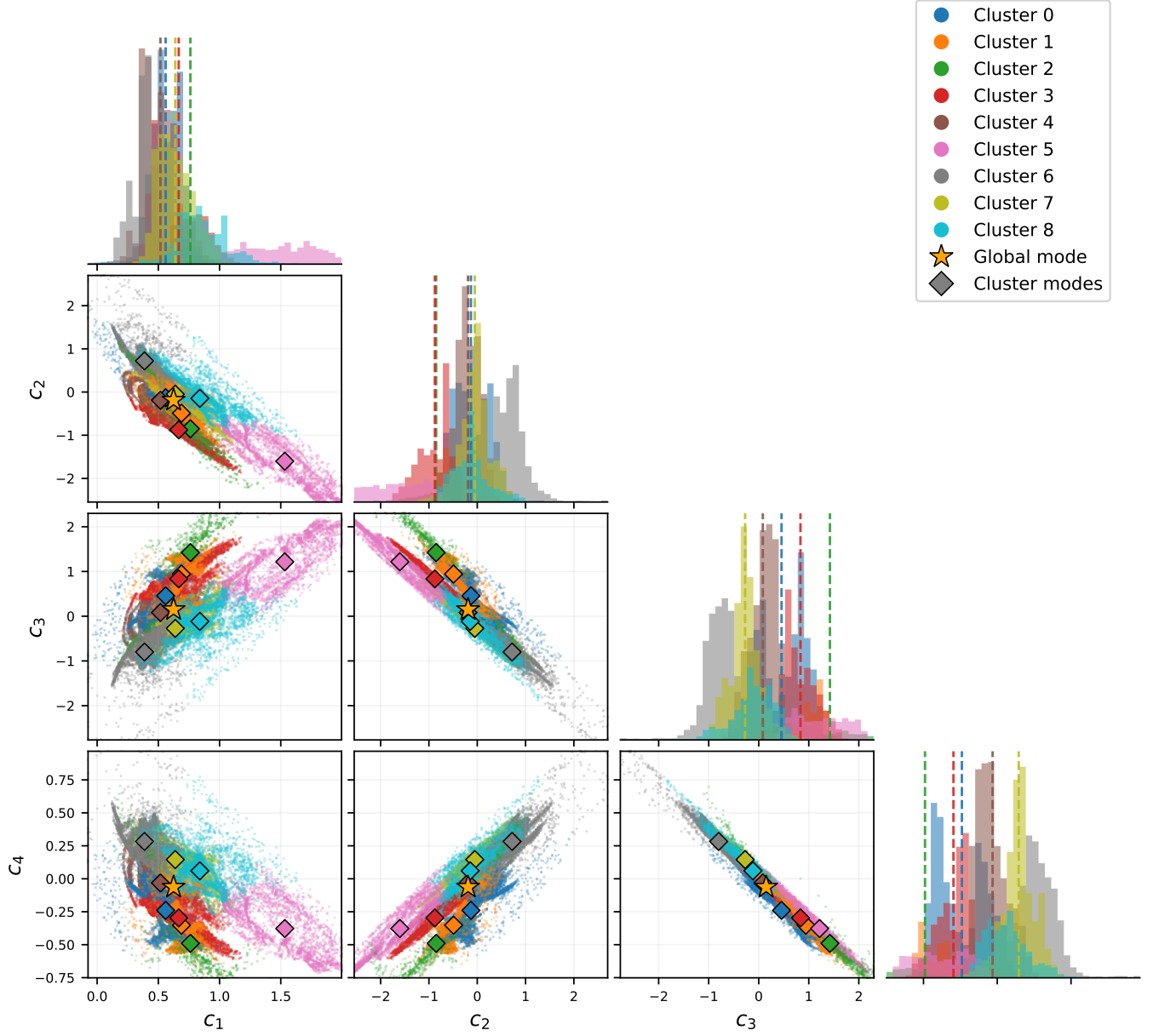


Figure C3. Hierarchical clustering of the LDC space. Corner plot of the four LDCs (c_1, c_2, c_3, c_4) colored by cluster membership as determined by the hierarchical clustering algorithm described in [Section 4.2](#). Each cluster is assigned a distinct color, with cluster modes shown as diamonds in the 2D panels. The global mode is depicted as an orange star. The diagonal panels show 1D marginal histograms of each coefficient, colored by cluster membership. Dashed vertical lines mark the modes for clusters 1, 3, and 4, which were used as injection–retrieval test cases in [Figure 5](#) and whose intensity profiles are shown in [Figure 3](#).

D. SENSITIVITY χ^2 MAPS

Figure D4 presents the results of our sensitivity analysis.

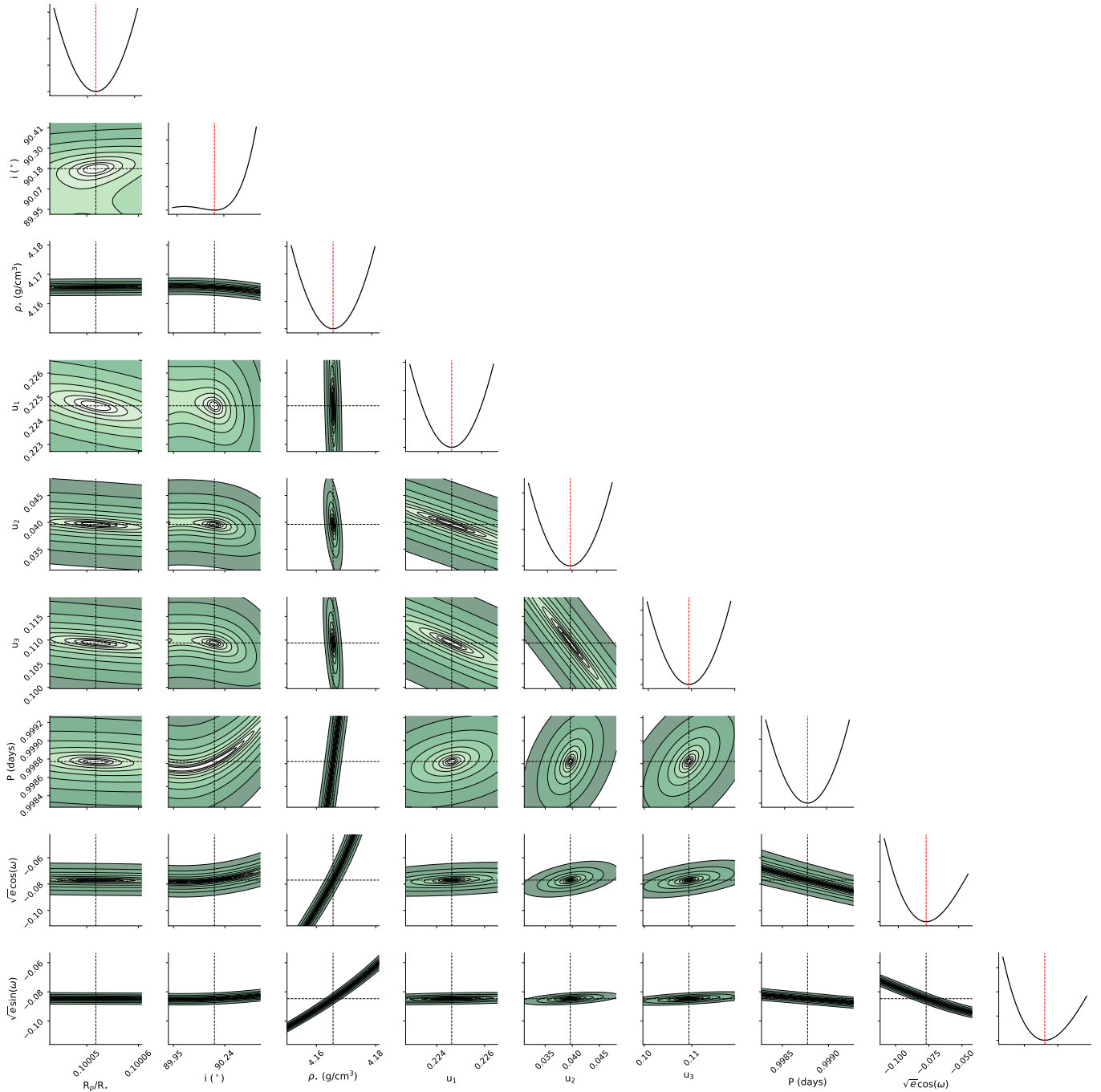


Figure D4. χ^2 maps across parameter pairs. Corner plot of the χ^2 maps used in our sensitivity test. Each diagonal subplot shows a 1-dimensional evaluation of the chi-squared space for the parameter corresponding to that column, where the vertical dashed red line indicates the truth value of that parameter. Off-diagonal subplots display two-dimensional iso- χ^2 contours derived from the χ^2 maps, where each map was obtained by varying the parameters indicated on the x and y axes, and the dashed black lines mark the truth values of the respective parameters. Finally, for certain contours that are not elliptical (e.g., $\sqrt{e} \sin \omega$ versus P) we approximate them as first-order polynomials and compute the rotation angle using $\theta = \arctan(m)$, where m is the slope.

E. COMPARISON OF MPS-ATLAS SET1 AND SET2 LDCs

997

998 Figure E5 summarizes the comparison of LDCs produced by the MPS-ATLAS set1 and set2 models and used in
 999 Section 6.1 to establish the strength of priors that can be placed on LDCs from state-of-the-art stellar models.

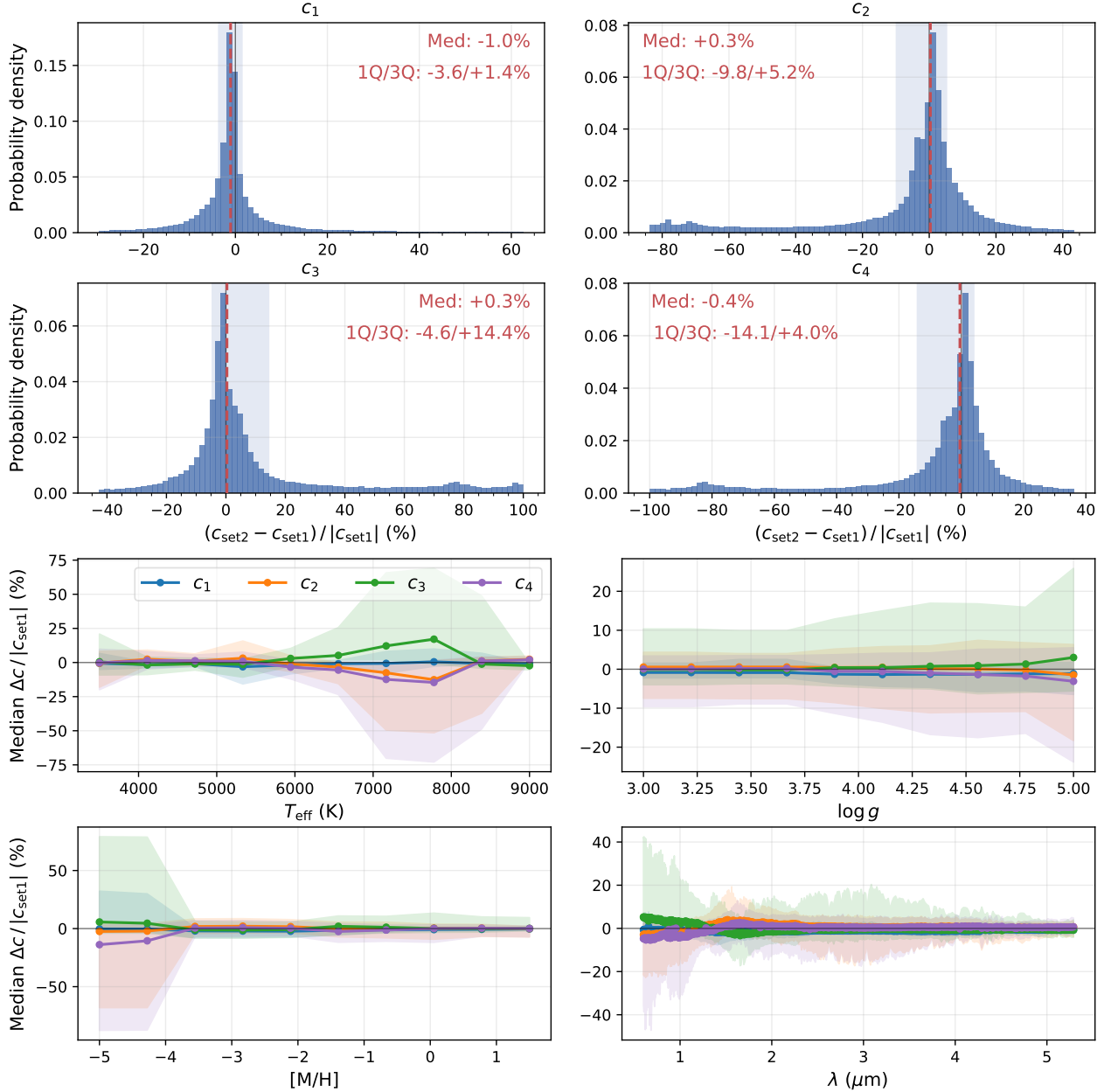


Figure E5. Comparison of LDCs between the MPS-ATLAS set1 and set2 stellar atmosphere models. *Top four:* Histograms of the signed relative difference $(c_{\text{set2}} - c_{\text{set1}}) / |c_{\text{set1}}|$ for each of the four non-linear LDL coefficients, computed point-by-point across matched grids of T_{eff} , $\log g$, $[M/H]$, and λ . Median values (red dashed lines) are near zero for all coefficients, confirming the negligible offset between the two model sets. The IQR grows with coefficient order, from 5% for c_1 to 15% for c_2 and 18–19% for c_3 and c_4 , reflecting the increasing difficulty in constraining higher-order coefficients that capture fine-scale intensity variations. *Bottom four:* Median relative difference as a function of stellar parameters and wavelength, with shaded regions indicating the IQR. The inter-model spread exhibits a clear dependence on the physical parameters considered, underscoring that the IQR derived from the top panels (and used to motivate the LDC priors in Section 6.1) are summary statistics that do not capture the the full breadth of behavior across the parameter space.